# Compact Face Representation via Forward Model Selection

Weiyuan Shao, Hong Wang, Yingbin Zheng, Hao Ye*

Shanghai Advanced Research Institute, Chinese Academy of Sciences

**Abstract.** This paper proposes a compact face representation for face recognition. The face with landmark points in the image is detected and then used to generate transformed face regions. Different types of regions form the transformed face region datasets, and face networks are trained. A novel forward model selection algorithm is designed to simultaneously select the complementary face models and generate the compact representation. Employing a public dataset as training set and fusing by only six selected face networks, the recognition system with this compact face representation achieves 99.05% accuracy on LFW benchmark.

**Keywords:** Face recognition, model selection, deep learning

## 1 Introduction

Face recognition in unconstrained environment is a challenge problem in computer vision research. Various hand-crafted representations such as LBP [1], Gabor [2], fisher faces [3] and eigenfaces [4] were designed to train the face recognition models. However, there still exists a gap between the recognition results and human visual system. Recently, Krizhevsky et al. [5] proposed to train the deep convolutional neural network (CNN) for image classification. A significant amount of progress has gone into recognizing faces via the deep networks and achieved human-level performance [6–14].

Previous deep learning based face recognition systems usually use two directions when generating the representations. One approach is to train a very deep network, e.g., the GoogLeNet style Inception with more than 20 layers [10], and obtain a well-defined representation of human faces. Another is to train several medium scale networks and then combine their output features to form a compound feature, e.g., [7, 6, 8]. In this paper, we present a compact face representation based on the forward model selection of deep networks for recognition task. The networks are trained from face dataset with machine labeled landmark points; each of them are corresponding to a specific face region and scale. We develop the forward model selection algorithm to select the complementary network set and dimensional reduction is finally applied on the concatenation of selected features to generate a compact representation.

* Corresponding author, E-mail: yeh@sari.ac.cn

The highlights of this work are: (i) compared with previous approaches such as [9], smaller number of networks are needed for the final representation; (ii) we present the analysis on different face regions and investigate the contribution of the regions on the model combination; (iii) as trained using a published available dataset, our compact face representation outperforms previous approaches based on the same dataset, and achieve comparable result with recognition systems from private datasets.

The rest of this paper is organized as follows. Section 2 discusses related works on face recognition. Our approach is described in Section 3 and evaluated on LFW dataset in Section 4. Finally, Section 5 concludes this paper.

## 2 Related Works

There have been a number of works focusing on recognizing faces in images. A few papers investigated the problem by using one deep network. For example, [10] presented the FaceNet system by employing more than 200 million face images for training and achieved state-of-the-art. However, this large-scale dataset is not publicly available. Based on the relatively small datasets, a popular strategy is to embed multiple networks. In [6], the DeepFace system was proposed with four stages, i.e., detection, alignment, representing and classification. It employed explicit 3D face modeling to apply a piecewise affine transformation on the original face images and generated new aligned face images. Three 8-layer networks are trained on the images using different alignments and color channels, and then combined by various protocols. The DeepFace was extended by the DeepID system[8], which trained 25 small deep convolutional networks from fixed regions and formed the representation by using the last hidden layer of each network. The successive research along this direction focused on improving the network structure [9]. Contrast to the previous approaches, our method chooses regions around the landmark points of human face as input, and performs forward model selection and dimensional reduction on networks to get compact representation. Figure 1 shows our face recognition pipeline.

## 3 Compact Face Representation

### 3.1 Pre-Processing

The first step of our pipeline is to detect the faces. The face detector from the open source software dlib [15] is used. We choose this off-the-shelf solution as the focus of this work is on the recognition task. However, we believe that our method is complementary to state-of-the-art face detection methods such as cascaded CNN [16] since they perform on different steps of the whole framework.

Through a network with the structure of [5], the corresponding facial landmark points in each face are obtained. Consider the time complexity, we select in-plane 2D alignment method which only use 5 landmark points (i.e., nose, left eye, right eye, left mouth, and right mouth) to align the face image. Then we can

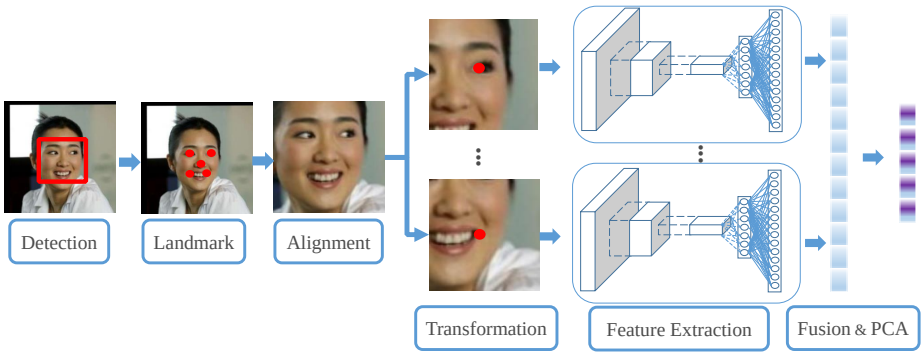


Fig. 1: Face recognition pipeline. We first detect the face region in the given image. The facial landmark points are calculated and used to align the face. Then we apply several transformation strategies, each of which is applied on the aligned face to generate the corresponding transformed face. The transformed face datasets are formed by grouping different kind of transformed faces and we use each of the datasets to train individual networks. The features are extracted from the fully connected layers in each network. Finally, forward model selection and dimensional reduction are applied on the features and the generated representation is used for face recognition task.

Table 1: Face network configuration.

| Layer name | conv1 | conv2_x | conv3_x | conv4_x | conv5_x | fc1 | fc2 | prob |
|---|---|---|---|---|---|---|---|---|
| Output size | 128×128 | 64×64 | 32×32 | 16×16 | 8×8 | 512 | 10,575 | 10,575 |
| Patch size, #channel | 5×5, 96 | 1×1, 96<br>3×3, 192 | 1×1, 192<br>3×3, 384 | 1×1, 384<br>3×3, 256 | 1×1, 256<br>3×3, 256 | - | - | - |

align and transform the faces based on the landmark points to weaken the influence from the pose variation and augment the versatility of faces. The transform matrix is shown in Equation (1), where $s$ controls the distance between two eyes, $\theta$ makes two eyes horizontal, and ($dx$,$dy$) makes the transformed face centered by a certain landmark point.

$$T = \begin{bmatrix} 1 & 0 & \frac{w}{2} \\ 0 & 1 & \frac{h}{2} \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} s & 0 & 0 \\ 0 & s & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos(\theta) & -\sin(\theta) & 0 \\ \sin(\theta) & \cos(\theta) & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & dx \\ 0 & 1 & dy \\ 0 & 0 & 1 \end{bmatrix} \tag{1}$$

### 3.2 Face Networks

We use a few deep networks to classify unique individuals. The input of the network is $X = \{(x_1, y_1), (x_2, y_2), ..., (x_n, y_n)\}$, where $x_i$ is the face image after transformation and $y_i$ is the corresponding identity. We follow the network architecture of lighted CNN [12], which consists of 9 convolutional layers (conv) and 2 fully connected layers (fc). Different from traditional CNN, here the active

**Algorithm 1** Forward Model Selection

```
procedure ForwardModelSelection(CandidateSet)
    SelectedSet ← ∅
    PreviousAccuracy ← 0.0
    for all Feature ∈ CandidateSet do
        WorkSet ← SelectedSet ⋃ {Feature}
        WorkRepresentation ← PCA(Concat(WorkSet))
        CurrentAccuracy ← Accuracy(Representation)
        if CurrentAccuracy > PreviousAccuracy then
            PreviousAccuracy ← CurrentAccuracy
            SelectedSet ← WorkSet
            SelectedRepresentation ← WorkRepresentation
        end if
    end for
    return (SelectedSet, SelectedRepresentation)
end procedure
```

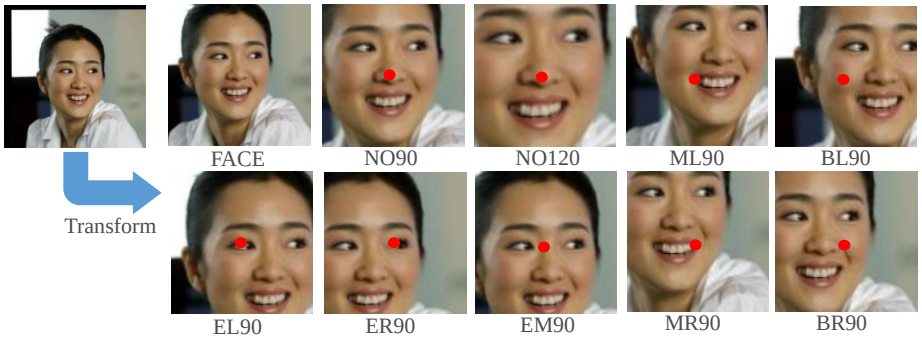


Fig. 2: Example of transformed face and partial regions.

function is Maxout [17], which outperforms the ReLU function in both our evaluation and previous face recognition work [12]. The detail network configuration is listed in Table 1.

As mentioned in previous works (e.g., [8]), if the network can discriminate large number of individuals, the feature from the fully connected layer is a good representation in face recognition task, and the recognition problem can be transformed to classification problem. We choose cross entropy loss $entropyLoss = -\sum_{i=1}^{n} y_i \log \hat{p}_i$ as the supervisory signal, where $\hat{p}_i$ is the predicted probability of the $i$th identity. Since multiple face region datasets based on different affine transformation are generated in previous sub-section, we denote the features set as $F = \{f_1, f_2, ..., f_k\}$, where $k$ is the amount of datasets.

Table 2: Comparison of networks from different transformed face datasets.

| Model | FACE | ER90 | EL90 | NO90 | MR90 | ML90 |
|---|---|---|---|---|---|---|
| Accuracy | 98.26% | 98.09% | 98.04% | 97.92% | 97.18% | 96.75% |

### 3.3 Forward Model Selection

Traditionally, face features are obtained by concatenate feature vectors in set $F$ [8]. In this subsection, we will introduce a forward model selection algorithm on $k$ kinds of feature vectors. Each feature vector is corresponding to a specific transformed face region and with different representing ability. For example, we found that regions centered by the nose are more discriminant than that centered by mouth corners in our experiments. However, simply selecting the best models and concatenating them directly may not get a good performance, since some models are not complementary with others. On the other hand, traversing all the subset of $F$ to find the optimal solution is not practical, since the search space will grow exponentially when the size of $F$ increases. Therefore, we propose a greedy forward model selection algorithm to find the near optimal solution; the pseudocode is detailed in Algorithm 1.

Initially, we set the Candidate Set $C = F$. In each iteration, one model will be extracted from $C$ and added to $S$. The selected one can mostly improve the previous performance from $S$. Principal Component Analysis (PCA) is applied on the concatenation of selected features. This step not only plays a dimensionality reduction role, but also drops the noise that may influent the recognition performance. We observe that most elements of the selected set are the models trained from the regions near eyes and nose, and the final performance is influenced by the traverse order. Different traverse order strategies will be discussed in next section. Finally, we use cosine distance as the similarity metric on the compact face representation for face recognition.

## 4 Experiments

### 4.1 Setup

**Dataset.** We evaluate our method in the Labeled Faces in the Wild (LFW) benchmark [18], which is the most popular benchmark for face recognition task. The dataset contains 13,233 face images with 5,749 individuals, and our experiments follow the unrestricted settings protocol which allows using external data in the training period. The benchmark randomly generates 6,000 face pairs and uses 10-fold cross-validation to avoid unfairly overfitting. Our models are trained using CASIA-WebFace dataset [7]. It contains 49,414 images with 10,575 individuals, and is a large public available face recognition dataset.

**Pre-processing.** In our experiments, several anchor points are defined based on facial landmark points, i.e., left eye (EL), right eye (ER), middle of two eyes (EM), below left eye (BL), below right eye (BR), nose (NO), left mouth corner (ML) and right mouth corner (MR). The face images are scaled according

Table 3: Comparison with different model traverse orders.

| Model Traverse Order | Selected Models | Accuracy |
|---|---|---|
| FACE,ER90,EL90,NO90,MR90,ML90 | FACE,ER90,EL90 | 98.66% |
| NO90,FACE,ER90,EL90,ML90,MR90 | NO90,FACE,ER90,EL90,MR90 | 98.61% |
| ML90,MR90,EL90,ER90,NO90,FACE | ML90,MR90,EL90,ER90,NO90,FACE | 98.54% |

Table 4: Performance of forward model selection for different PCA compression rates.

| Dimensions | w/o PCA | 256 | 128 | 64 | 32 | 16 |
|---|---|---|---|---|---|---|
| Accuracy | 98.71% | 99.00% | **99.05%** | 98.97% | 98.26% | 96.81% |

to the distance between two eyes with 90 or 120 pixels. Then different images centered by each landmark point are cropped with the size of $144 \times 144$ pixels, and form the transformed face datasets for the further training of different face recognition network. To discriminate each dataset and the corresponding model, we use FACE as the model trained based on the whole detected faces, and {AnchorPoint + DistanceBetweenTwoEyes} to nominate the network from the partial region. For example, the network trained by the transformed face centered by nose and scaled the distance between two eyes to 90 pixels is nominated as NO90. An example of the regions are illustrated in Figure 2.

**Network training.** In order to overcome the overfitting, the dropout is used after fully connected layer with ratio 0.75. Our mini-batch is 40 and the momentum is set to 0.9. The weight decay is set to be 0.0005 and 0.005 in convolutional layer and fully connected layer, respectively. The weights in each layer are initialized by xavier strategy. The learning rate is set to 0.001 and each network is trained on an Nvidia Titan X GPU for about 3 days.

### 4.2 Comparisons

**Transformed face regions.** We start by evaluating models from different transformed face datasets. Regions near the 5 facial landmark points are selected to find their importance and the result is illustrated in Table 2. FACE model obtains the highest performance, as they contain all of the face information. The ER and EL models get 98% accuracy, while ML and MR have around 97% accuracy. It denotes that the transformed faces fetched from the upper part of face may be more discriminant than these from lower part in face recognition task. The reason may be that the variation in the lower part of face is larger than the upper part in the unconstraint environment, which will confuse the network.

**Model traverse orders.** The forward model selection solution is highly dependent on the traverse order. As shown in Table 3, we find that if we traverse under the decreasing order based on the performance[1] of different models (row 1 in Table 3), the performance after fusing will be higher than random order

[1] The performance is the cross-validation result on CASIA-WebFace, which has a same order with the result of LFW.

Table 5: Comparison with previous works using CASIA-WebFace for training.

| Approach | #Nets | Accuracy |
|---|---|---|
| WebFace [7] | 1 | 97.73% |
| Lightened CNN [12] | 1 | 98.13% |
| Face Search [13] | 7 | 98.23% |
| MM-DFR-JB [14] | 8 | 99.02% |
| Ours | 6 | 99.05% |

Table 6: Comparison with state-of-the-art approaches with different training datasets.

| Approach | Dataset | Available | People | Images | Accuracy |
|---|---|---|---|---|---|
| DeepFace[6] | SFC | private | 4,030 | 4.4M | 97.35% |
| VGG Net [11] | VGG Face | public | 2,622 | 2.6M | 98.95% |
| DeepID2+[9] | CelebFaces+ | private | 10,177 | 202K | 99.47% |
| FaceNet[10] | Google | private | 8M | 200M | 99.63% |
| Ours | CASIA-WebFace | public | 10,575 | 500K | 99.05% |

(row 2 in Table 3) and increasing order (row 3 in Table 3). Another observation is that fusing more number of models may not improve the results.

**Forward model selection.** We construct a larger candidate set including the FACE model and all combinations of the 8 anchor points (see Figure 2) and 2 scales (90 and 120 pixels). After running the forward model selection algorithm, a selected set is generated with 6 networks, i.e., {EM90, FACE, ER90, EL90, NO90, NO120}. We evaluate the performance of different versions of forward model selection without PCA or after PCA compression to a different number of dimensions (Table 4). PCA works surprisingly well; the forward model selection can be compressed to 128 or even to 64 dimensions almost without any quality loss. Overall, the forward model selection boosts the performance from 98.26% (FACE model only) to 99.05% (128 dimension), and meanwhile keeps the face representation compact.

**Comparison with state-of-the-art.** Table 5 and Table 6 compare our method with the state-of-the-art methods in LFW. Our result is better than the methods trained based on CASIA-WebFace. And the performance reaches the same magnitude of methods trained using other public or private datasets.

## 5 Conclusion

In this paper, we have introduced a compact face representation based on the forward model selection of deep networks for face recognition. After setting up the pre-processing methods, given an image, the face with landmark points is detected and then used to generate transformed face regions. Several face networks are trained using the transformed face region datasets. The forward model selection algorithm is designed to simultaneously select the complementary face models and generate the compact representation. Experiments on the LFW benchmark demonstrate the effectiveness of our compact face representation.

**Acknowledgments.** We would like to thank the anonymous reviewers for their helpful comments. This work was supported by the STCSM's Program (No. 16511104802).

## References

1. Ahonen, T., Hadid, A., Pietikainen, M.: Face description with local binary patterns: Application to face recognition. IEEE Transaction on Pattern Analysis and Machine Intelligence **28**(12) (2006) 2037–2041
2. Liu, C., Wechsler, H.: Gabor feature based classification using the enhanced fisher linear discriminant model for face recognition. IEEE Transaction on Image Processing **11**(4) (2002)
3. Guillaumin, M., Verbeek, J., Schmid, C.: Is that you? metric learning approaches for face identification. In: ICCV. (2009) 498–505
4. Turk, M.A., Pentland, A.P.: Face recognition using eigenfaces. In: CVPR. (1991)
5. Krizhevsky, A., Sutskever, I., Hinton, G.E.: Imagenet classification with deep convolutional neural networks. In: NIPS. (2012) 1097–1105
6. Taigman, Y., Yang, M., Ranzato, M., Wolf, L.: Deepface: Closing the gap to human-level performance in face verification. In: CVPR. (2014) 1701–1708
7. Yi, D., Lei, Z., Liao, S., Li, S.Z.: Learning face representation from scratch. arXiv preprint arXiv:1411.7923 (2014)
8. Sun, Y., Wang, X., Tang, X.: Deep learning face representation from predicting 10,000 classes. In: CVPR. (2014) 1891–1898
9. Sun, Y., Wang, X., Tang, X.: Deeply learned face representations are sparse, selective, and robust. In: CVPR. (2015) 2892–2900
10. Schroff, F., Kalenichenko, D., Philbin, J.: Facenet: A unified embedding for face recognition and clustering. In: CVPR. (2015) 815–823
11. Parkhi, O.M., Vedaldi, A., Zisserman, A.: Deep face recognition. In: BMVC. (2015)
12. Wu, X., He, R., Sun, Z.: A lightened cnn for deep face representation. arXiv preprint arXiv:1511.02683 (2015)
13. Wang, D., Otto, C., Jain, A.K.: Face search at scale: 80 million gallery. arXiv preprint arXiv:1507.07242 (2015)
14. Ding, C., Tao, D.: Robust face recognition via multimodal deep face representation. IEEE Transaction on Multimedia **17**(11) (2015) 2049–2058
15. King, D.E.: Dlib-ml: A machine learning toolkit. Journal of Machine Learning Research **10** (2009) 1755–1758
16. Li, H., Lin, Z., Shen, X., Brandt, J., Hua, G.: A convolutional neural network cascade for face detection. In: CVPR. (2015) 5325–5334
17. Goodfellow, I.J., Warde-Farley, D., Mirza, M., Courville, A., Bengio, Y.: Maxout networks. arXiv preprint arXiv:1302.4389 (2013)
18. Huang, G.B., Ramesh, M., Berg, T., Learned-Miller, E.: Labeled faces in the wild: A database for studying face recognition in unconstrained environments. Technical report, Technical Report 07-49, University of Massachusetts, Amherst (2007)